Manan Agrawal

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SUMMARY

AI Engineer specializing in agentic AI systems, LLM infrastructure, and intelligent developer tooling. Experienced in building production-grade generative AI platforms, multi-agent workflows, RAG pipelines, and scalable ML systems using Python, AWS, LangChain, MLflow, and FastAPI. Built AI-driven automation systems integrating CI/CD, infrastructure tooling, code generation, and real time inference pipelines processing 500K+ records daily.

WORK EXPERIENCE

Martinrea International US Inc. **2025 - 2026**

Research and Development AI Intern

- Designed and deployed agentic AI workflows integrating LLMs, orchestration pipelines, and code-generation-assisted development to automate enterprise software delivery and AI-driven operational workflows.
- Built scalable ML pipelines covering ingestion, feature engineering, inference, evaluation, and monitoring with integrated CI/CD workflows processing 500K+ records daily.
- Developed distributed backend AI services integrating APIs, vector databases, and infrastructure automation for real-time intelligent workflows.
- Improved system performance by 35% and reduced latency by 25% through optimization, experimentation, and scalable system design.

Honeywell **Jan 2025 - May 2025**

GenAI Project Intern

- Designed and deployed an AWS-based LLM email classification system processing 1,000+ requests/day with 98%+ accuracy while reducing manual effort by 60%.
- Built RAG pipelines with embedding-based semantic retrieval and tool-assisted reasoning workflows using historical enterprise data.
- Developed hierarchical classification and confidence-scoring systems with human-in-the-loop validation and automated monitoring pipelines.
- Built scalable inference pipelines integrating APIs, vector databases, CI/CD workflows, and enterprise monitoring systems.

SS Infotech **Oct 2023 - Jun 2024**

Machine Learning Engineering Intern

- Developed "CV Robo," an ML-powered resume analysis platform using NLP, semantic matching, and similarity scoring to improve ATS compatibility for job applications.
- Built keyword extraction, recommendation, and ranking pipelines to compare resumes against job descriptions and identify optimization opportunities.
- Implemented scalable backend pipelines and REST APIs using Python and SQL for automated parsing, scoring, and workflow automation.

TECHNICAL SKILLS

- **Languages**: Python, Java, C++, SQL, JavaScript
- **Frameworks**: PyTorch, TensorFlow, FastAPI, LangChain, MLflow, LlamaIndex
- **AI Systems**: Generative AI, LLMs, Agentic AI, Multi-Agent Systems, RAG, Tool Use Agents, Code Generation
- **Data & MLOps**: SQL, FAISS, Pinecone, CI/CD, Model Monitoring, AWS, Docker, Kubernetes
- **Systems**: Distributed Systems, REST APIs, Async Programming, Scalable Infrastructure

PROJECTS

LLM Based Document Understanding and Retrieval System

- Built production-ready semantic search and AI-powered document retrieval pipelines using transformer embeddings and FAISS, deploying services on AWS with automated CI/CD for sub-second contextual retrieval across enterprise datasets.
- Designed conversational AI workflows integrating LLMs, vector search, and retrieval-augmented generation to extract structured insights from unstructured documents.
- Developed scalable API-driven services for intelligent document understanding, recommendation generation, and enterprise knowledge discovery applications, containerized with Docker and deployed on AWS ECS.

Multi Agent AI Decision Intelligence System (Thesis)

- Developed a multi-agent machine learning framework combining forecasting, sentiment analysis, and probabilistic reasoning to optimize large-scale financial decision workflows using Python, XGBoost, LSTM, and statistical modeling on AWS SageMaker.
- Designed experimentation and backtesting pipelines on 50+ years of historical data with automated evaluation, risk modeling, and agent-based feedback loops for adaptive decision optimization.
- Implemented scalable ML workflows integrating real-time data processing, model orchestration with MLflow, and performance monitoring via automated CI/CD pipelines for high-throughput predictive analytics.

HONORS AND PUBLICATIONS

- IEEE Peer Reviewer for machine learning and AI conferences and journals
- Published 15+ research papers in machine learning, AI, and applied statistics
- Govt. of India Design Patent No 386153 001
- Winner, Most Complete Solution, LPL Financial Hackathon (Alpha: AI platform)
- Invited Guest Speaker, Johnson C. Smith University, Charlotte (AI and Finance)

EDUCATION

UNC Charlotte **Aug 2024 - May 2026**

MS, Computer Engineering

RCOEM, Nagpur University **Jul 2020 - Jun 2024**

BTech, Electronics Engineering